

Lecture 4: First derivatives and convex function

MA3236 Non-Linear Programming

Hu Yukuan

yukuan.hu@nus.edu.sg
S17-08-21

Quick review

- Convex optimization is special yet useful, possessing nice theoretical properties (shown in **this!** and later lectures :).
- Convex sets contain all possible line segments.

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C, \quad \forall \mathbf{x}, \mathbf{y} \in C, \lambda \in [0, 1].$$

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- Convex functions satisfy “convex inequalities”.

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in C, \lambda \in [0, 1].$$

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Nonnegative linear combinations, pointwise maxima, special compositions preserve convexity.

- Convex function $\Leftrightarrow$ its epigraph is convex.

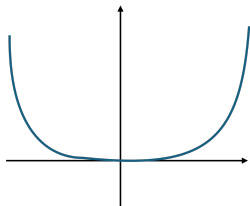
Theorem 4.1 (No gap in convex optimization)

Let C be a nonempty **convex subset** of $\mathbf{R}^n$, and $f : C \rightarrow \mathbf{R}$ is a **convex function**. Consider the convex minimization problem:

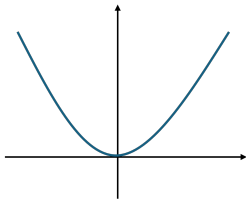
$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in C. \end{array}$$

Suppose $\mathbf{x}^* \in C$ is a local minimizer to the problem. Then

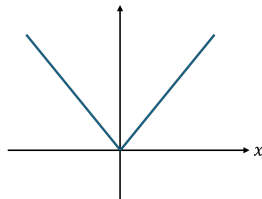
- 1 $\mathbf{x}^*$ is a global minimizer.
- 2 if f is strictly convex, then $\mathbf{x}^*$ is the **unique** global minimizer.



f convex, but not strictly convex

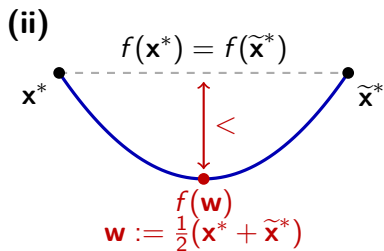
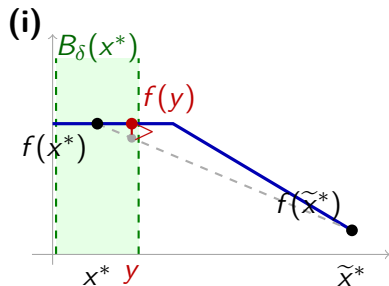


f (strictly) convex



f convex, but not strictly convex

Geometry behind the proof



For convex minimization problem (f convex function, C convex set):

$$\mathbf{x}^* \text{ is a local minimizer} \Rightarrow \begin{cases} \mathbf{x}^* \text{ is a global minimizer} \\ f \text{ is strictly convex} \Rightarrow \mathbf{x}^* \text{ is the unique global minimizer} \end{cases}$$

Proof. (i) Suppose on the contrary that $\mathbf{x}^*$ is not a global minimizer. Then there exists a better feasible point $\tilde{\mathbf{x}}^* \in C$ such that $f(\tilde{\mathbf{x}}^*) < f(\mathbf{x}^*)$. Since C is convex, $[\mathbf{x}^*, \tilde{\mathbf{x}}^*] \subseteq C$.

Since $\mathbf{x}^*$ is a local minimizer, choose $\delta \in (0, \|\tilde{\mathbf{x}}^* - \mathbf{x}^*\|)$ such that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in C \text{ with } \|\mathbf{x} - \mathbf{x}^*\| < \delta.$$

Define

$$\theta := \frac{\delta}{2\|\tilde{\mathbf{x}}^* - \mathbf{x}^*\|} \in \left(0, \frac{1}{2}\right), \quad \mathbf{y} := (1 - \theta)\mathbf{x}^* + \theta\tilde{\mathbf{x}}^* \in C.$$

Then $\|\mathbf{y} - \mathbf{x}^*\| = \delta/2 < \delta$, so $f(\mathbf{y}) \geq f(\mathbf{x}^*)$. But convexity gives

$$f(\mathbf{y}) \leq (1 - \theta)f(\mathbf{x}^*) + \theta f(\tilde{\mathbf{x}}^*) < f(\mathbf{x}^*),$$

which leads to a contradiction.

For convex minimization problem (f convex function, C convex set):

$$\mathbf{x}^* \text{ is a local minimizer} \Rightarrow \begin{cases} \mathbf{x}^* \text{ is a global minimizer} \\ f \text{ is strictly convex} \Rightarrow \mathbf{x}^* \text{ is the unique global minimizer} \end{cases}$$

Proof (cont'd). (ii) Suppose on the contrary there is another global minimizer $\tilde{\mathbf{x}}^* \neq \mathbf{x}^*$ with the same global minimum value $f(\tilde{\mathbf{x}}^*) = f(\mathbf{x}^*)$.

Consider $\mathbf{w} = \frac{1}{2}(\mathbf{x}^* + \tilde{\mathbf{x}}^*) \in C$. Then

$$f(\mathbf{w}) = f\left(\frac{1}{2}(\mathbf{x}^* + \tilde{\mathbf{x}}^*)\right) < \frac{1}{2}f(\mathbf{x}^*) + \frac{1}{2}f(\tilde{\mathbf{x}}^*) = f(\mathbf{x}^*),$$

contradicting the fact that $\mathbf{x}^*$ is a global minimizer. □

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Question. If a convex function has a unique global minimizer, is it strictly convex?

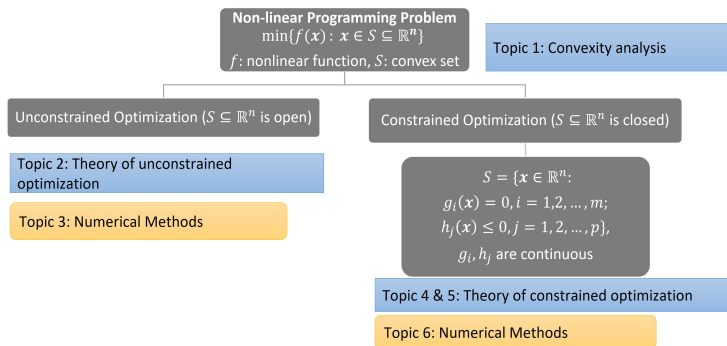
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Topics



In lectures 3–5, we will study convex analysis (Topic 1).

- Convex sets and convex functions.
- First-order characterizations of convex functions.
- Second-order characterizations of convex functions.

Differentiable functions

Definition 4.2 (Interior point)

Let $S \subseteq \mathbf{R}^n$ be a nonempty set. A point $\mathbf{x}_0 \in S$ is called an **interior point** of S if there exists $\epsilon > 0$ such that

$$\|\mathbf{y} - \mathbf{x}_0\| < \epsilon \implies \mathbf{y} \in S, \quad \forall \mathbf{y} \in \mathbf{R}^n.$$

Definition 4.3 (Gradient vector)

Let $S \subseteq \mathbf{R}^n$, let $f : S \rightarrow \mathbf{R}$, and let $\mathbf{x}$ be an interior point of S at which f is differentiable. The **gradient vector** of f at $\mathbf{x}$ is

$$\nabla f(\mathbf{x}) := \begin{pmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{pmatrix}.$$

Example 4.4 (Linear function)

Let $\mathbf{b} \in \mathbf{R}^n$ and define $f : \mathbf{R}^n \rightarrow \mathbf{R}$ by $f(\mathbf{x}) := \mathbf{b}^T \mathbf{x}$. Find $\nabla f(\mathbf{x})$.

Example 4.4 (Linear function)

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Solution. Rewrite f as

$$f(\mathbf{x}) = \sum_{i=1}^n b_i x_i.$$

Therefore,

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) = b_i, \quad \forall \mathbf{x} \in \mathbf{R}^n, \quad i = 1, \dots, n.$$

Collecting these results gives

$$\nabla f(\mathbf{x}) = \mathbf{b}, \quad \forall \mathbf{x} \in \mathbf{R}^n.$$

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Question. For $A \in \mathbf{R}^{n \times n}$, how about the quadratic function

$$q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^T A \mathbf{x}?$$

Useful results on $\nabla f(\mathbf{x})$

Proposition 4.5 (Directional derivatives and curves)

Let $S \subseteq \mathbf{R}^n$, let $f : S \rightarrow \mathbf{R}$, and suppose that f is differentiable at an interior point $\mathbf{x} \in S$.

(a) For every $\mathbf{d} \in \mathbf{R}^n$,

$$\nabla f(\mathbf{x})^T \mathbf{d} = \lim_{\lambda \rightarrow 0} \frac{f(\mathbf{x} + \lambda \mathbf{d}) - f(\mathbf{x})}{\lambda}.$$

(b) Let $\delta > 0$ and let $\gamma : (-\delta, \delta) \rightarrow S$ be differentiable at 0 with $\gamma(0) = \mathbf{x}$. Define $g : (-\delta, \delta) \rightarrow \mathbf{R}$ by $g := f \circ \gamma$. Then

$$g'(0) = \nabla f(\mathbf{x})^T \gamma'(0).$$

Useful results on $\nabla f(\mathbf{x})$ (cont'd)

Proposition 4.5 (cont'd)

- (c) Suppose $\nabla f(\mathbf{x}) \neq \mathbf{0}$. Among unit directions, f decreases most rapidly along

$$\mathbf{d}_{\text{sd}} := -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|},$$

and increases most rapidly along $-\mathbf{d}_{\text{sd}}$.

For every unit vector $\mathbf{d}$,

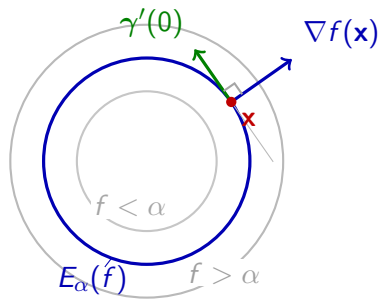
$$-\|\nabla f(\mathbf{x})\| \leq \nabla f(\mathbf{x})^T \mathbf{d} \leq \|\nabla f(\mathbf{x})\|.$$

If $\nabla f(\mathbf{x}) = \mathbf{0}$, every directional derivative is zero.

Useful results on $\nabla f(\mathbf{x})$ (cont'd)

Let $D \subseteq \mathbf{R}^n$, let $f : D \rightarrow \mathbf{R}$, and let $\alpha \in \mathbf{R}$. The **level set** of f at level α is

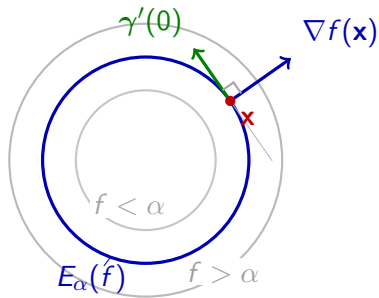
$$E_\alpha(f) := \{\mathbf{x} \in D : f(\mathbf{x}) = \alpha\}.$$



Useful results on $\nabla f(\mathbf{x})$ (cont'd)

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Proposition 4.6 (Gradient normality)

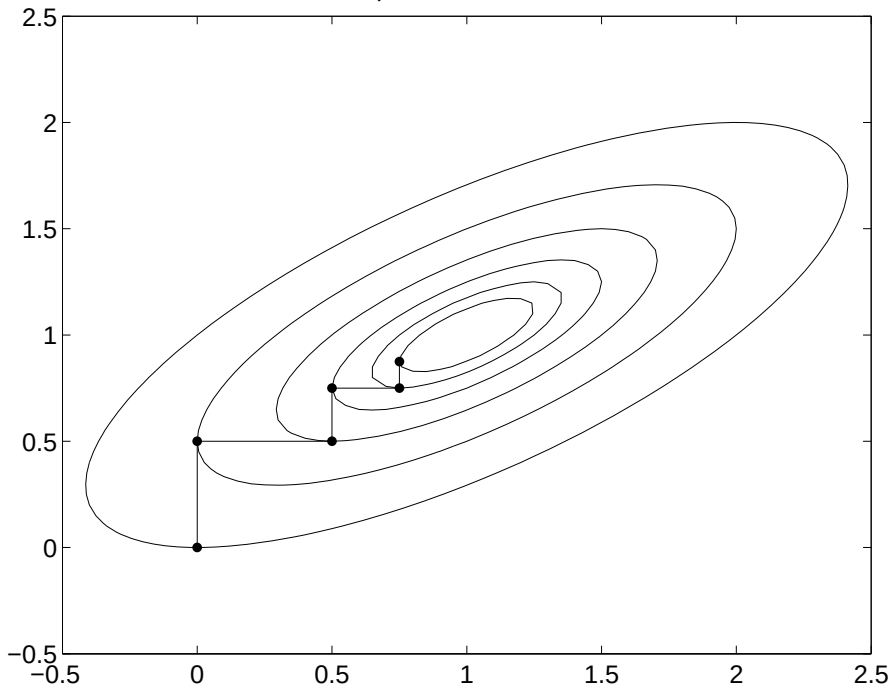
Let $\mathbf{x} \in E_\alpha(f)$ be an interior point of D , and suppose that f is differentiable at $\mathbf{x}$ with $\nabla f(\mathbf{x}) \neq \mathbf{0}$.

For $\delta > 0$, let $\gamma : (-\delta, \delta) \rightarrow E_\alpha(f)$ be differentiable at 0 with $\gamma(0) = \mathbf{x}$. Then

$$\nabla f(\mathbf{x})^T \gamma'(0) = 0.$$

Warning. Do not confuse the level set $E_\alpha(f)$ with the sublevel set $L_\alpha(f) := \{\mathbf{x} \in D : f(\mathbf{x}) \leq \alpha\}$.

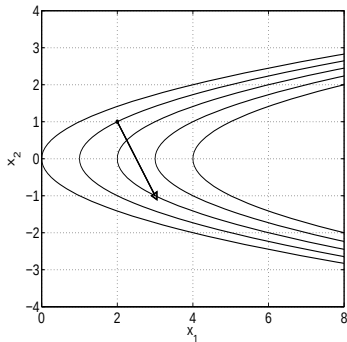
Steepest descent iterates



Example 4.7

Define $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ by $f(x_1, x_2) := x_1 - x_2^2$. Then $\nabla f(\mathbf{x}) = \begin{pmatrix} 1 \\ -2x_2 \end{pmatrix}$.

Consider the level curve $f(x_1, x_2) = x_1 - x_2^2 = 1$.

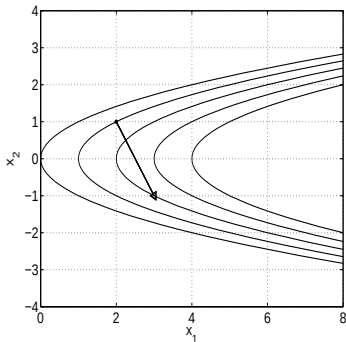


- Contours of $f(\mathbf{x}) = x_1 - x_2^2$. Starting from the left, the contour values are 0, 1, 2, 3, 4.

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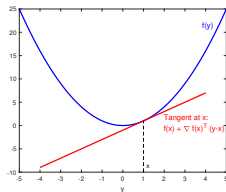
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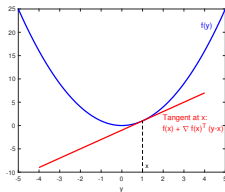
- Contours of $f(\mathbf{x}) = x_1 - x_2^2$.
Starting from the left, the contour values are 0, 1, 2, 3, 4.
- At $\mathbf{x}^* := [2; 1]$, $\nabla f(\mathbf{x}^*) = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$,
which points in the direction of increasing $f(\mathbf{x})$ (i.e. $f(\mathbf{x}) = \alpha$ where α is increasing)

Tangent plane characterization of convex function



The tangent plane is **below** the graph of a convex function.

Tangent plane characterization of convex function



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Theorem 4.8 (Tangent plane characterization of convex functions)

Let $C \subseteq \mathbf{R}^n$ be open and convex, and let $f : C \rightarrow \mathbf{R}$ have continuous first partial derivatives. Then

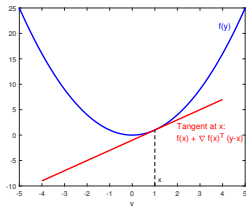
(a) the function f is convex if and only if

$$f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) \quad \forall \mathbf{x}, \mathbf{y} \in C.$$

(b) the function f is strictly convex if and only if

$$f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) < f(\mathbf{y}) \quad \forall \mathbf{x}, \mathbf{y} \in C, \mathbf{x} \neq \mathbf{y}.$$

Tangent plane characterization of convex function (cont'd)



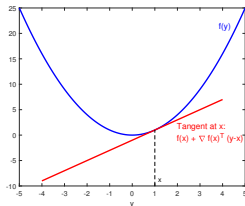
The tangent plane is **below** the graph of a convex function.

Equivalent characterization for convex f :

$$\begin{aligned} f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) &\leq f(\mathbf{y}) \\ \iff f(\mathbf{x}) + \nabla f(\mathbf{y})^T (\mathbf{y} - \mathbf{x}) &\geq f(\mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in C. \end{aligned}$$

For strictly convex f and $\mathbf{x} \neq \mathbf{y}$, replace $\leq, \geq$ by $<, >$.

Tangent plane characterization of convex function (cont'd)



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For strictly convex f and $\mathbf{x} \neq \mathbf{y}$, replace $\leq, \geq$ by $<, >$.

Question. What if the inequality fails for some pair of distinct points?

Example 4.9

Define $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ by $f(x_1, x_2) := x_1^2 + x_2^2$. Show that f is convex.

Solution. $\nabla f(x_1, x_2) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}$.

$$\begin{aligned} f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) &= x_1^2 + x_2^2 + [2x_1, 2x_2] \begin{pmatrix} y_1 - x_1 \\ y_2 - x_2 \end{pmatrix} \\ &= 2x_1y_1 - x_1^2 + 2x_2y_2 - x_2^2 = -(x_1 - y_1)^2 + y_1^2 - (x_2 - y_2)^2 + y_2^2 \\ &\leq y_1^2 + y_2^2 = f(\mathbf{y}). \end{aligned}$$

Optimality condition for a convex optimization

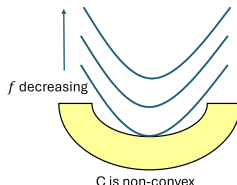
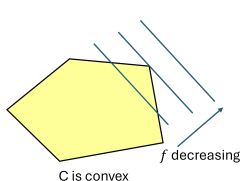
Theorem 4.10

Let $U \subseteq \mathbf{R}^n$ be open and convex, let $C \subseteq U$ be nonempty and convex, and let $f : U \rightarrow \mathbf{R}$ be convex and continuously differentiable. Then $\mathbf{x}^* \in C$ is a global minimizer of the minimization problem

$$\min\{f(\mathbf{x}) \mid \mathbf{x} \in C\}.$$

if and only if

$$\nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) \geq 0, \quad \forall \mathbf{x} \in C.$$



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For any $\mathbf{x} \in C$ sufficiently close to $\mathbf{x}^*$,

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) + o(\|\mathbf{x} - \mathbf{x}^*\|)$$

Proof of Theorem 4.10 “ $\Leftarrow$ ”

Suppose $\nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq 0$ for all $\mathbf{x} \in C$. Then

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq f(\mathbf{x}^*) \quad \forall \mathbf{x} \in C.$$

Hence $\mathbf{x}^*$ is a global minimizer.

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“ $\Rightarrow$ ”

Conversely, suppose $\mathbf{x}^*$ is a global minimizer. Fix $\bar{\mathbf{x}} \in C$ and define

$$\phi(t) := f(\mathbf{x}^* + t(\bar{\mathbf{x}} - \mathbf{x}^*)), \quad t \in [0, 1].$$

Convexity of C makes this segment feasible, so $\phi(t) \geq \phi(0)$.

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Convexity of C makes this segment feasible, so $\phi(t) \geq \phi(0)$. Therefore,

$$0 \leq \lim_{t \downarrow 0} \frac{\phi(t) - \phi(0)}{t} = \nabla f(\mathbf{x}^*)^T(\bar{\mathbf{x}} - \mathbf{x}^*).$$

Since $\bar{\mathbf{x}}$ was arbitrary, the required inequality holds. □

Optimality condition for unconstrained convex optimization

Let $C \subseteq \mathbf{R}^n$ be **open** and convex, and let $f : C \rightarrow \mathbf{R}$ be convex and continuously differentiable. Then $\mathbf{x}^* \in C$ is a global minimizer of the minimization problem

$$\min\{f(\mathbf{x}) \mid \mathbf{x} \in C\}.$$

if and only if

$$\nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) \geq 0, \quad \forall \mathbf{x} \in C.$$

Choose $\delta > 0$ such that $B_\delta(\mathbf{x}^*) \subseteq C$. For every nonzero $\mathbf{d} \in \mathbf{R}^n$, both $\mathbf{x}^* \pm \frac{\delta}{2\|\mathbf{d}\|}\mathbf{d}$ lie in C . Applying the variational inequality to these two gives

$$\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0, \quad \nabla f(\mathbf{x}^*)^T (-\mathbf{d}) \geq 0.$$

Hence $\nabla f(\mathbf{x}^*)^T \mathbf{d} = 0$ for every $\mathbf{d}$, so $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

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Theorem 4.11

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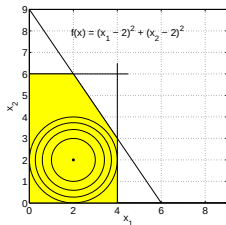
$$\min\{f(\mathbf{x}) \mid \mathbf{x} \in C\}.$$

if and only if $\nabla f(\mathbf{x}^) = 0$.*

Recall Examples 2.1 and 2.2. Compute $\nabla f(\mathbf{x}^*)$.

Example 2.2

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 2)^2 \\ \text{subject to} & x_1 \leq 4 \\ & x_2 \leq 6 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0.\end{array}$$

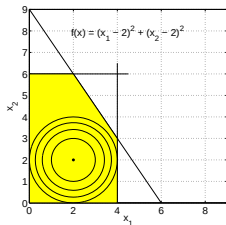


- $\mathbf{x}^* = [2; 2]$ in the interior of feasible region
- $\nabla f(\mathbf{x}) = \begin{pmatrix} 2(x_1 - 2) \\ 2(x_2 - 2) \end{pmatrix}$
- $\nabla f(\mathbf{x}^*) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Recall Examples 2.1 and 2.2. Compute $\nabla f(\mathbf{x}^*)$.

Example 2.2

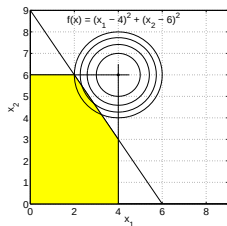
$$\begin{array}{lll} \text{minimize} & f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 2)^2 \\ \text{subject to} & x_1 \leq 4 \\ & x_2 \leq 6 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0. \end{array}$$



- $\mathbf{x}^* = [2; 2]$ in the interior of feasible region
- $\nabla f(\mathbf{x}) = \begin{pmatrix} 2(x_1 - 2) \\ 2(x_2 - 2) \end{pmatrix}$
- $\nabla f(\mathbf{x}^*) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Example 2.1

$$\begin{array}{lll} \text{minimize} & f(\mathbf{x}) = (x_1 - 4)^2 + (x_2 - 6)^2 \\ \text{subject to} & x_1 \leq 4 \\ & x_2 \leq 6 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0. \end{array}$$



- $\mathbf{x}^* = \left[\frac{34}{13}; \frac{66}{13} \right]$ on the boundary of feasible region
- $\nabla f(\mathbf{x}) = \begin{pmatrix} 2(x_1 - 4) \\ 2(x_2 - 6) \end{pmatrix}$
- $\nabla f(\mathbf{x}^*) = \begin{pmatrix} -\frac{36}{13} \\ -\frac{24}{13} \end{pmatrix} \neq \mathbf{0}$

Proof of Theorem 4.8 (a) f is convex $\Leftrightarrow f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}), \forall \mathbf{x}, \mathbf{y} \in C$.

Proof. (a) “ $\Rightarrow$ ”

Suppose f is convex on C . Let $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in (0, 1]$. Then

$$\begin{aligned} f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) &= f(\lambda\mathbf{y} + (1 - \lambda)\mathbf{x}) \leq \lambda f(\mathbf{y}) + (1 - \lambda)f(\mathbf{x}) \\ \Rightarrow \quad \frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} &\leq f(\mathbf{y}) - f(\mathbf{x}) \end{aligned}$$

Thus,

$$\lim_{\lambda \downarrow 0} \frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x})$$

But

$$\lim_{\lambda \downarrow 0} \frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} = \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$$

Therefore, we have

$$\nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x}).$$

Proof of Theorem 4.8 (a) f is convex $\Leftrightarrow f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}), \forall \mathbf{x}, \mathbf{y} \in C$.

Proof. (a) “ $\Leftarrow$ ”

Conversely, suppose

$$f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) \quad \forall \mathbf{x}, \mathbf{y} \in C. \quad (1)$$

holds. Let $\mathbf{u}, \mathbf{v} \in C$, $\lambda \in (0, 1)$ and define $\mathbf{w} := \lambda \mathbf{u} + (1 - \lambda)\mathbf{v}$. Then

$$\mathbf{u} - \mathbf{w} = (1 - \lambda)(\mathbf{u} - \mathbf{v}), \quad \mathbf{v} - \mathbf{w} = -\lambda(\mathbf{u} - \mathbf{v}).$$

Applying (1) to the pairs $\mathbf{w}, \mathbf{u}$ and $\mathbf{w}, \mathbf{v}$ gives

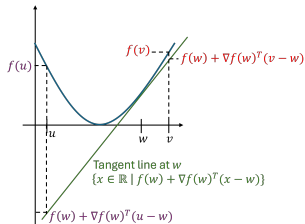
$$f(\mathbf{w}) + \nabla f(\mathbf{w})^T(\mathbf{u} - \mathbf{w}) \leq f(\mathbf{u}), \quad (2)$$

$$f(\mathbf{w}) + \nabla f(\mathbf{w})^T(\mathbf{v} - \mathbf{w}) \leq f(\mathbf{v}) \quad (3)$$

Now $(2) \times \lambda + (3) \times (1 - \lambda)$ gives

$$f(\mathbf{w}) \leq \lambda f(\mathbf{u}) + (1 - \lambda)f(\mathbf{v})$$

$$\Rightarrow f(\lambda \mathbf{u} + (1 - \lambda)\mathbf{v}) \leq \lambda f(\mathbf{u}) + (1 - \lambda)f(\mathbf{v}).$$



Proof of Theorem 4.8 (b) f is strictly convex $\Leftrightarrow f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) < f(\mathbf{y}), \forall \mathbf{x}, \mathbf{y} \in C$.

Proof. (b) “ $\Leftarrow$ ”

If the strict tangent-plane inequality holds, then the two endpoint inequalities used in the proof of (a) are strict, which gives strict convexity.

“ $\Rightarrow$ ”

By part (a), the non-strict tangent-plane inequality already holds. If strictness failed for distinct points, there would exist $\mathbf{x} \neq \mathbf{y} \in C$ such that

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$$

By strict convexity of f , we have

$$\begin{aligned} \frac{1}{2}f(\mathbf{x}) + \frac{1}{2}f(\mathbf{y}) &> f\left(\frac{1}{2}\mathbf{x} + \frac{1}{2}\mathbf{y}\right) = f\left(\mathbf{x} + \frac{1}{2}(\mathbf{y} - \mathbf{x})\right) \\ &\geq f(\mathbf{x}) + \frac{1}{2}\nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) = \frac{1}{2}f(\mathbf{x}) + \frac{1}{2}f(\mathbf{y}) \end{aligned}$$

which is a contradiction.

Take-home messages

- For convex optimization, local minimizer $\Leftrightarrow$ global minimizer; strict convexity gives uniqueness when a minimizer exists.
- On an open convex domain, a continuously differentiable function is convex $\Leftrightarrow$ every tangent plane globally underestimates it.
- For a differentiable convex objective over a convex feasible set, the variational inequality characterizes global minimizers.
- Questions:
 - other analytical ways to verify convexity **without comparing?**
 - optimality conditions for **constrained** convex optimization?